Notes on Bayesian Adaptive Platform Design

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2024-01-23

Summary

The focus of this document is to describe an optimization scheme to search for the best BOP2 Design (Zhou et al. 2017). A brief background on the BOP2 design is as follows: the endpoints of the phase II trials can be represented by a random variable $Y \sim \text{Multinom}(\theta_1, \ldots, \theta_K)$, where $\theta_k = P(Y = k)$ is the probability that Y belongs to the kth category. Given a total sample size of N patients, the BOP2 design consists of $R-1$ interim looks, occurring when the number of enrolled patients reaches $n_1, \ldots, n_{R-1}$, and a final look when $n_R = N$ patients are enrolled. At the rth interim look, let $D_r = (x_1, \ldots, x_K)$ denote the interim data, and x_k denote the number of patients with $Y = k$. Note that $\sum_{k=1}^{K} x_k = n_r$ patients. If the data show an undesired result, the trial can be stopped earlier. The stopping rule is defined as $P(\mathbf{b}\boldsymbol{\theta} \leq \phi \mid D_r) > C(n_r)$, where $\mathbf{b} = (b_1, \ldots, b_K)$ is a pre-specified vector with elements 0 and 1, and ϕ is a pre-specified threshold. The probability cutoff $C(n_r)$ is determined by optimizing the statistical power according to the null and alternative hypotheses, and an assumed type I error rate.

Figure 1 shows the optimization scheme of the BOP2 design. The user inputs the following parameters:

1. the type of the trail,
2. the null hypothesis, as the threshold ϕ of the stopping rule,
3. the alternative hypothesis, for power calculation,
4. the type I error rate.

Then, the BOP2 optimization problem in general can be a three-layer optimization problem through the formula

$$\max_{N,R} \left\{ \max_{n_1 < n_2 < \cdots < n_{R-1}} \left\{ \max_{\lambda,\gamma} \text{Power}(\lambda, \gamma) \right\} \right\} \tag{1}$$

The first layer is to decide the total sample size, N, and the number of interim looks, R. The second layer is to decide the sample sizes that the interim looks occur, $n_1, \ldots, n_{R-1}$ (Variable 1 in Figure 1). The third layer is to decide the The probability cutoffs $C(n_r) = 1 - \lambda(n_r/N)^\gamma$ for $r = 1, \ldots, R$ (Variable 2 in Figure 1). **Currently, we can focus on the optimization methods for the second and the third layers of (1), and, suppose N and R are fixed.** The pseudo-code of BOP2 design optimization is

- Given ϕ_0s for H_0 and ϕ_1s for H_1, and, desired type I error rate.
- Given N: maximum sample size, and, R: times of interim looks.
- Optimize $n_1 < n_2 < \cdots < n_{R-1}$: the number of enrolled patients at each interim look (Layer 2).
  1. Initialize $n_1^{(0)} < n_2^{(0)} < \cdots < n_{R-1}^{(0)}$.
  2. At iteration t,
  a. Optimize power $\text{Power}(\lambda, \gamma)$ based on $n_r^{(t-1)}$ for $r = 1, \ldots, R$ (Layer 3).
  b. Update $n_1^{(t)} < n_2^{(t)} < \cdots < n_{R-1}^{(t)}$.
  c. Repeat a and b until the stopping criterion is met.
- Return the best $n_1^* < n_2^* < \cdots < n_{R-1}^*$ and the corresponding probability cutoffs (the stopping boundary) and the power.

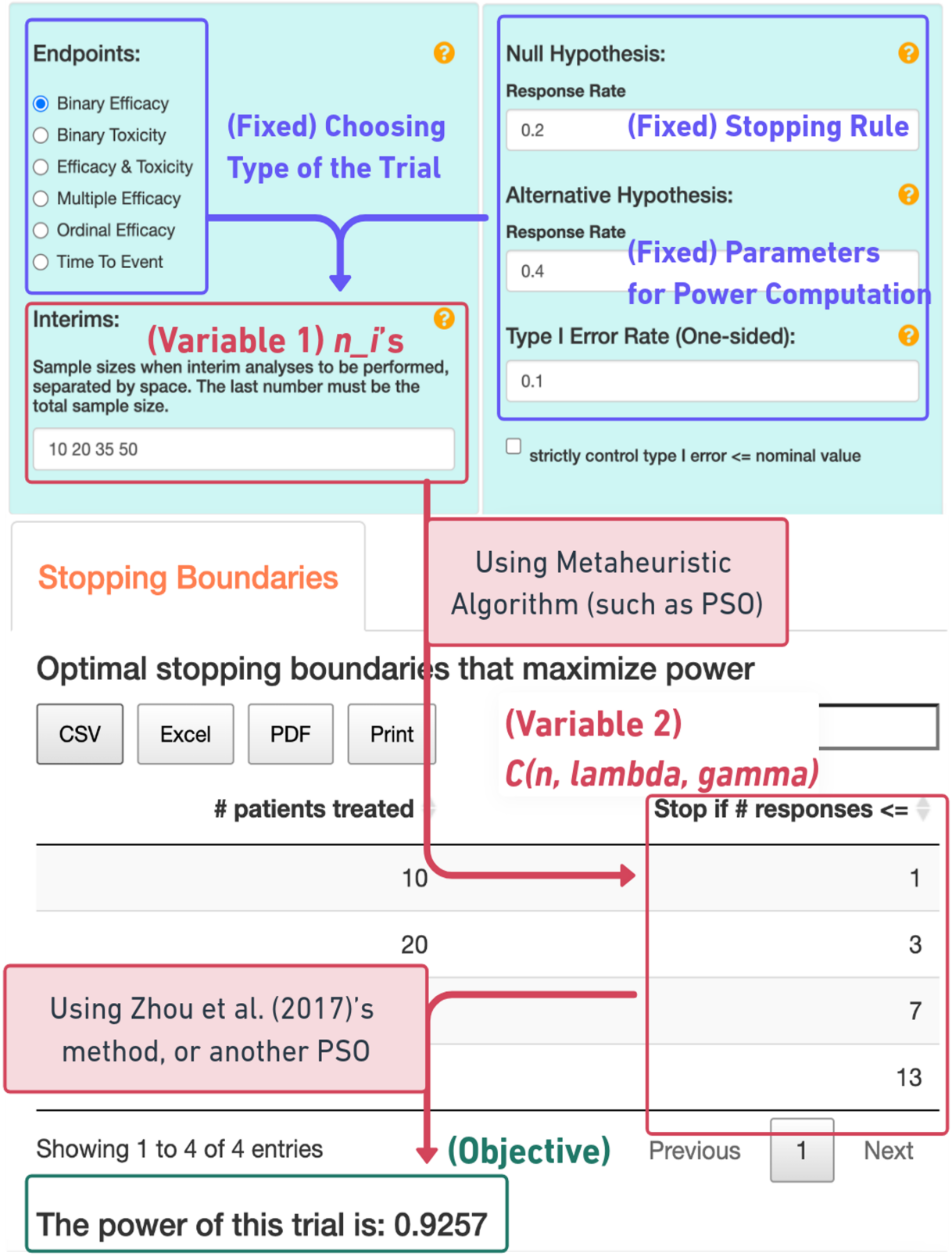


Figure 1: The flow of optimizing a BOP2 design.

The challenges of this project are:

- The search space of $(n_1, n_2, \ldots, n_{R-1})$ is a discrete space with constraint $n_1 < n_2 < \cdots < n_{R-1} < N$. However, the original PSO was proposed to search on the continuous space. To properly search for the BOP2 design, we need to modify the original PSO or to try some other existing algorithms that fit the search requirements.
- To conduct the numerical experiment for optimizing (1), we need more computational details on the third layer loop, optimizing $C(n_r)$ for each interim look, in (Zhou et al. 2017). The information we need includes the search space of λ and γ, the statistical power computation at each interim look, and, the statistical power computation shown on the final resulting table on the website https://trialdesign.org/ (refer to the bottom of Figure 1).

Online PSO Code

The PSO Codes are available on the GitHub repository https://github.com/PingYangChen/globpso. The system requirements are as follows:

- OS: Windows 11
- R version: >= 4.3.2
- Rtools version: 4.3 https://cran.r-project.org/bin/windows/Rtools/.
  - Post-installation, ensure that the path `C:\YOUR INSTALLATION PATH\rtools43\mingw64\bin` is included in the PATH environment variable.
- R package Dependency
  - Rcpp >= 1.0.12
  - RcppArmadillo >= 0.12.6.6.1

When the system requirement are satisfied, one can install the `globpso` package via R package `remote`.

```
# Install globpso using remote
install.packages("remote")
remotes::install_github("PingYangChen/globpso")
```

To use the PSO code, the following is the example code. This example shows how to define the objective function, with non-variable miscellaneous information, and to run the PSO algorithm

```
library(globpso)
# The objective function is the 3-dimensional quadratic function with "location shift"
objf <- function(x, loc) {
 val <- 0
 for (i in 1:length(x)) val <- val + (x[i] - loc)^2
 return(val)
}
# Define the search domain to be [-5, 5]^3 (3-dimensional variable)
low_bound <- rep(-5, 3); upp_bound <- rep(5, 3)
# Define the non-variable miscellaneous information, "location shift", of objf to be 1
loc_shift <- 1
# Run PSO for this optimization problem, also input for the misc info, loc = loc_shift
res <- globpso(objFunc = objf, lower = low_bound, upper = upp_bound, loc = loc_shift)
res$par # The optimized variable
res$val # The optimized objective function value
```

First Try on Feb 2024

- Fix total sample size N and number of interim looks R
- Since the basic PSO only works on the continuous domain, we can treat PSO particles as a weight vector of length $R that

$$\mathbf{w} = (w_1, w_2, \ldots, w_{N_c}), \quad \sum w_j = 1$$

- Then, the final sample size of the j-th cohort is $[N * w_j]$.

Let $N = 50$ and $R = 4$ be fixed, the PSO finds the best 4-stage cohort is 38 44 48 50 with power 0.9671 using 32 particles and 30 iterations.

Optimal stopping boundaries that maximize power

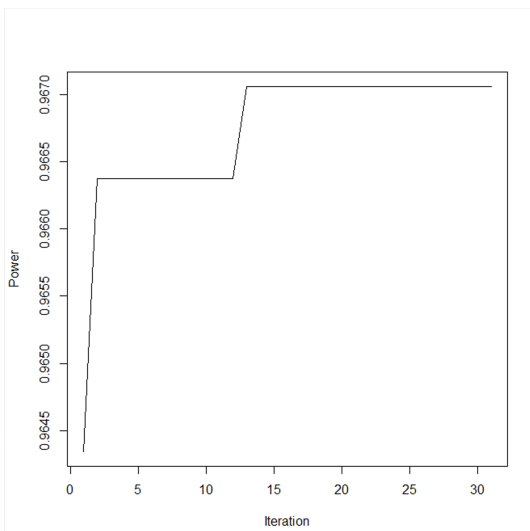
CSV Excel PDF Print Search:

# patients treated	Stop if # responses <=
38	8
44	10
48	12
50	13

Showing 1 to 4 of 4 entries Previous 1 Next

The power of this trial is: 0.9671

The figure below is the power increment during the PSO search over 30 iterations.



Discussions:

- If only consider optimizing power, the results may not be clinically useful. For example, in the above result, the first interim look is too late.
  - It needs to be more careful on the constraint of the sample sizes of the interim looks.
  - Possible solutions are
    - Restricting the first interim look should be stopped at 20% to 50% of the total sample size.
    - Using a compound utility function that considers power and the sample size simultaneously.

References

Zhou H, Lee JJ, Yuan Y (2017) BOP2: Bayesian optimal design for phase II clinical trials with simple and complex endpoints. Statistics in medicine 36:3302–3314

Appendix: Background Knowledge of the BOP2 Design

Examples of Scenarios

1. **Binary Efficacy Endpoint:**

- Primary endpoint is the objective response rate (ORR)
- Futile if ORR$\leq$ 20%, Promising if ORR$\geq$ 40%
- $Y = 1$, objective responses; $Y = 0$, o.w.

2. **Nested Efficacy Endpoint:**

- Complete remission, Partial remission, Stable disease, Progressive disease
- CR and PR are favorable
- Promising if $P(\text{CR} \cup \text{PR}) > 30\%$ and $P(\text{CR}) > 15\%$.
- $Y = 1, 2, 3, 4$ for CR, PR, SD, PD, respectively.

3. **Co-primary Efficacy Endpoint:**

- Two co-primary endpoints, ORR and event-free survival at 6 months (EFS6)
- Futile if ORR$\leq$ 10% and EFS6$\leq$ 20%
- $Y = 1$ for (OR, EFS6); $Y = 2$ for (OR, no EFS6); $Y = 3$ for (no OR, EFS6); $Y = 4$ for (no OR, no EFS6).

4. **Jointly Monitoring Efficacy and Toxicity:**

- Primary endpoint is the objective response rate (ORR)
- Monitors Toxicity Rate (TR)
- Futile if ORR$\leq$ 45% and TR$>$ 30%
- $Y = 1$ for (Toxicity, OR); $Y = 2$ for (no Toxicity, OR); $Y = 3$ for (Toxicity, no OR); $Y = 4$ for (no Toxicity, no OR).

Probability Models

All the scenarios can be unified by representing the response variable Y that follows a multinomial distribution,

$$Y \sim \text{Multinom}(\theta_1, \ldots, \theta_K)$$

where $\theta_k = P(Y = k)$ is the prob. that Y belongs to the kth category, $k = 1, \ldots, K$.

Suppose a total of n patients have been enrolled into the trial and their endpoints are evaluated. Let $D_n = (x_1, \ldots, x_k)$ denote the interim data and x_k denote the number of patients with Y_k, where $\sum x_k = n$.

Assume $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_K)^\top$ follows Dirichlet prior

$$(\theta_1, \ldots, \theta_K) \sim \text{Dir}(a_1, \ldots, a_K)$$

where $a_1, \ldots, a_K$ are hyperparameters. The posterior distribution of $\boldsymbol{\theta}$ is also a Dirichlet distribution that

$$\boldsymbol{\theta} \mid D_n \sim \text{Dir}(a_1 + x_k, \ldots, a_K + x_K).$$

For scenario 1, this DM model becomes a beta-binomail model.

Trail Design

Define

- N is the maximum sample size,
- There are R interim looks occurring when the number of enrolled patients reaches $n_1 < n_2 < \cdots < n_R < N$, and, a final look when all N patients are enrolled.
- $C(n) = 1 - \lambda(n/N)^\gamma$ is the probability cutoff.

- (S1) $P(\theta_1 \leq 0.2 \mid D_n) > C(n)$;
- (S2) $P(\theta_1 \leq 0.15 \mid D_n) > C(n)$ and $P(\theta_1 + \theta_3 \leq 0.3 \mid D_n) > C(n)$;
- (S3) $P(\theta_1 + \theta_2 \leq 0.1 \mid D_n) > C(n)$ and $P(\theta_1 + \theta_3 \leq 0.2 \mid D_n) > C(n)$;
- (S4) $P(\theta_1 + \theta_2 \leq 0.45 \mid D_n) > C(n)$ or $P(\theta_1 + \theta_3 > 0.3 \mid D_n) > C(n)$.

Property: Given $\boldsymbol{\theta} \sim \text{Dir}(a_1 + x_1, \ldots, a_K + x_K)$ and a design vector $\mathbf{b} = (b_1, \ldots, b_K)$ with elements 0 and 1, $\mathbf{b}\boldsymbol{\theta}$ follows a beta distribution

$$\text{Beta}\left(\sum_{k=1}^{K} b_k(a_k + x_k), \sum_{k=1}^{K}(1 - b_k)(a_k + x_k)\right).$$

As a result, $P(\mathbf{b}\boldsymbol{\theta} \leq \phi \mid D_n)$ can be easily evaluated as

$$P(\mathbf{b}\boldsymbol{\theta} \leq \phi \mid D_n) = B\left(\phi; \sum_{k=1}^{K} b_k(a_k + x_k), \sum_{k=1}^{K}(1 - b_k)(a_k + x_k)\right)$$

where $B(\phi; \zeta, \xi)$ is the cdf of a beta distribution with parameters ζ and ξ, evaluated at ϕ.

Lemma: $P(\mathbf{b}\boldsymbol{\theta} \leq \phi \mid D_n)$ is a monotonic function of $\sum_{k=1}^{K} b_k x_k$.

- The lemma's contribution: simply the stopping boundary definition.
  - Only have to count the number of relevant events and decision go/nogo based on whether the count exceeds the boundary.

Deciding the Probability Cutoff

Approach 1:

1. Claim H_0 and H_1, and, the desired type I error rate.
2. Find (λ, γ) yield the desired type I error rate by grid search.
3. Among those (λ, γ) candidates identified in Step 2, select the one that yields the maximum power.

Approach 2:

1. Claim H_0 and H_1, and, the desired type I and type II error rates.
2. Find (N, λ, γ) yield the desired type I and type II error rates by grid search.

- $N \in [N_{min}, N_{max}]$ where $N_{max} = N$.

3. Among those (N, λ, γ) candidates identified in Step 2, select the one that yields the smallest $E(N \mid H_0)$.

Layer 3 Optimization: BOP2 Generation

Given

- N: maximum sample size
- R: times of interim looks
- $n_1 < n_2 < \cdots < n_R < N$: the number of enrolled patients reaches at each interim look
- ϕ_0s for H_0 and ϕ_1s for H_1.
- Desired type I error rate

Optimize

- Parameter
  - λ and γ in $C(n)$
- Objective function

$$\max_{\lambda,\gamma} \text{Power}(\lambda, \gamma)$$

or for fixed type II error rate

$$\max_{N,\lambda,\gamma} \text{Power}(N, \lambda, \gamma)$$

Output

- $s_1 < s_2 < \cdots < s_R < s_N$: The stopping boundary at each interim look.
  - $s_i \leq n_i$, $i = 1, \ldots, R$ and $s_N \leq N$.
- Statistic Power